08E — Deep-Dive Feature Audit: Outside-Sect World & Ecology (Part 5 of 6)

Southern Spirit River Fishing, Spirit-Fish Cuisine, Underwater Chests, Wandering Jianghu Figures (`OUT-07, 20`), Campus Visitors & Legendary Clue Quests

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Autonomous Xianxia Sandbox**. River fishing, spirit-fish banquets, wandering sword-seekers, hermit doctors, visiting scholars, and multi-step legendary clue expeditions must interlock seamlessly with sect hunger, study speed, 43-Path subpaths, and autonomous NPC social life!
- **Master Roadmap for Category 08 (`08A–08F`)**:
  - `08A`: Macro Geopolitics, 19 Rival Sects, Diplomacy, Vassals, 6 Resource Nodes & Siege Warfare
  - `08B`: Expeditions, Caravans, Midnight Black Market, Tavern Bounties, Espionage & `v2.6.0` Living Chronicle
  - `08C`: Secret Realms, 43-Floor Nine-Dao Pagoda, 4 Sanctums, Puzzle Gates, 4 Micro-Locations & Immortal Tombs
  - `08D`: Wilderness Ecology, Roaming Beast Packs, Menagerie, Pets, Symbiosis & Wild Resource Gathering
  - **`08E` (This Document)**: **Southern Spirit River Fishing, Spirit Cuisine, Underwater Chests, Wandering Figures, Campus Visitors & Legendary Clue Quests (`fishing.js`, `visitors_clues.js`, `expedition.js`, `ai.js`)**
  - `08F`: Grand Finale — Holistic Overview of the Entire Outside-Sect World & Ecology

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1. `[FISH-SYS-01]` Five Concrete Bugs in Southern Spirit River Fishing & Spirit-Fish Cuisine (`js/fishing.js` & `js/ai.js`)!

1. **Starvation Next to a Kitchen Full of Cooked Fish Dishes (`A.eat.can` in `js/ai.js` vs `js/fishing.js`)!**:
   - `A.eat.can` in `js/ai.js` checks **`ECO.count(W, 'meal') > 0 || ECO.count(W, 'grain') > 0`—completely ignoring `W.fishing.cookedDishes` (`FISHING.hasCookedDish(W)`)!**

- If `meal === 0` and `grain === 0` during a winter grain shortage, **disciples starve to `0` satiety and die even when `25` freshly cooked Spirit-Fish Dishes are sitting ready in the canteen**!

2. **Double Meal Consumption & `+82.8` Double Satiety Bug (`A.eat.start` + `A.eat.tick`)**:
   - When `meal > 0` *and* an NPC eats a Spirit-Fish Dish, the fish dish hook grants immediate satiety (`+45..+75`) and consumes `1` fish dish, **while `A.eat` ALSO deducts `1` standard `'meal'` and ticks `+82.8` satiety** (`consuming 2 meals for 1 person and wasting half the satiety above 100`)!
   - **Fix**: When `A.eat.start` reserves a `fishDish`, set `ctx.ateFishDish = true` so `A.eat` does **not** deduct an extra `'meal'` unit and scales `tick` satiety cleanly without double-counting!
3. **Dead `studyMult` (`+12%` Scripture Study Speed) Fish-Dish Buff (`fishing.js` vs `ai.js` & `cpeco.js:383`)**:
   - Spirit-Fish dishes grant `studyMult` (`e.g. 1.12`), **which neither `A.study.tick` (`js/ai.js`) nor `A.guidedStudy.tick` (`js/cpeco.js:383`) ever reads (`0.00%` actual effect!)**! Wire `FISHING.getStudySpeedMult(n, W.t)` into both `A.study.tick` and `A.guidedStudy.tick`.
4. **Legacy 37-Subpath Table in Underwater Chest Drops & Missing `listedAtDay` (`js/fishing.js`)**:
   - Underwater Treasure Chests (งมหีบสมบัติใต้น้ำ) map subpath manuals using a hardcoded `37`-subpath tier table—forcing **all `~249` newer subpaths (`87%` of the game!) to Tier 1**—and mint chest equipment without `listedAtDay = W.day` (`causing 0-ms liquidation in EXT.sellStaleSectGear`, `04E`)! Derive subpath tier dynamically from `CULT.PATHS` / `CULT.trackDef` and stamp `listedAtDay: W.day`.
5. **Missing `TIDE_VOYAGER` (วิถีนาวิกสมุทร) & `SPIRIT_HUSBANDRY` Synergy**:
   - Grant `TIDE_VOYAGER` disciples `+25%` rare-fish / pearl-chest catch rate, higher `A.fish.score` affinity, and real `CULT.gain` (`tideInsight`) when catching Tier 3–5 Spirit Fish or defeating River Leviathans!

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2. `[CLUE-SYS-02] 6` Missing Legendary Weapon Clue Chains (`21 / 27` in `visitors_clues_weapons_suite.js:228`), `31` Missing Grand Dao Mythic Manual Clues, & Forced `'SWORD'` Skill Bug (`js/visitors_clues.js`)!

1. **Mathematical Proof From `visitors_clues_weapons_suite.js:27` vs `228`**:

- `VC.LEGENDARY_EQUIPMENT.length === 27` (`20` Weapons + `3` Armors + `4` Accessories), yet completing **all `60` `VC.CLUE_TEMPLATES` claims only `21` unique artifacts**!
- All **6 `v2.0.0` Grand Dao Legendary Weapons** (`LEG_WPN_BODHI_NIRVANA_ROSARY`, `LEG_WPN_TAOTIE_DEVOURING_CLAW`, `LEG_WPN_BAIHUA_SOUL_NEEDLE`, `LEG_WPN_BAJIAO_TEMPEST_FAN`, `LEG_WPN_HETU_LUOSHU_ASTROLABE`, `LEG_WPN_CAISHEN_GOLDEN_ABACUS`) declare `obtainSources: ['CLUE_QUEST']` with **zero `CLUE_TEMPLATES`**, making them **100% unobtainable in natural gameplay**!
- Likewise, the **31 `LEG_V230_<PATH>` Mythic Manuals** have `0` Visitor Clue chains (unlike the 31 legacy Legendaries)!
- **Fix**: Add 6 dedicated 3-step `CLUE_TEMPLATES` (`personal` skill/trait trials matching `ROSARY/meditation`, `CLAW/strength`, `NEEDLE/MEDICINE`, `FAN/tactics`, `ASTROLABE/ARRAY`, `ABACUS/numeracy`) + template generator/entries for `LEG_V230_<PATH>`!

2. **Guardian Encounter Forces `'SWORD'` Practice on Every Party Member (`js/visitors_clues.js:1798`)**:
   - In `VC.simulateGuardianEncounter` (`line 1798`), every surviving party member trains `'SWORD'` (`HC.practice(m, 'SWORD', ...)`) even if they wield a Spear, Fist, Bow, Zither, Brush, or Rosary! Replace hardcoded `'SWORD'` with `MARTIAL.primaryWeaponSkillKey(m)`!
3. **Closed-Loop Stone Verification on Ordinary Clues**:
   - Ensure any Spirit-Stone caches found via `CLUE_ORD_*` strictly debit `W.world.resourcePool.stones`.

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3. `[FIG-VIS-03]` Unfixed `OUT-07` (`HERMIT_DOCTOR` Free Bypass), Unfixed `OUT-20` (Duplicate Recruit), & `0%` Autonomous Interaction With `wanderingFigures` (`js/expedition.js:6370–6480`)!

1. **`OUT-07` — UI Hardcodes `{ free: true }` on `HERMIT_DOCTOR` (`js/ui.js:5958` & `expedition.js:6406–6423`)**:
   - `HERMIT_DOCTOR` defines a `120`-stone consultation fee, **yet `ui.js:5958` always passes `{ free: true }`**, granting instant free full-HP, `0`-toxicity, `0`-injury cures that bypass both `infirmary` and Tier-4 Meridian/Detox pills!
   - **Fix**: Remove `{ free: true }` from `ui.js` (`requiring 120 stones paid to W.world.resourcePool.stones`, or `free` only once if `affinity >= 65` or `charter === 'CHARTER_BROTHERHOOD_SANCTUARY'`), and enforce `1` patient per visit!

2. **OUT-20 — Duplicate `recruitWanderingFigure` Returns `{ ok: true }` (`expedition.js:6376, 6474`)**:
    - Check `if (fig.recruited || W.npcs.some(n => n._origFigureId === fig.id && !n.dead && !n.exiled))` **before** calling `hydrateFigureToNpc`, returning `{ ok: false, reason: 'already-recruited' }`!
3. **Zero Autonomous NPC Interaction With `W.world.wanderingFigures` & Enemy Spy Guests**:
    - Currently `W.world.wanderingFigures` (`SWORD_SEEKER`, `HERMIT_DOCTOR`, `ROGUE_ARRAYIST`, `UNORTHODOX_BANDIT`) never interact unless clicked in UI!
    - **Emergent Fix**:
        - When `SWORD_SEEKER` visits near the sect, let our `martial` Elder or ambitious `ARENA_GLORY` / `SWORD_IMMORTAL` disciples autonomously accept `1` `DUEL`!
        - When a critically injured Elder (`sev >= 2`) cannot be cured quickly and `W.sect.stones >= 120 + cashReserve`, let `healer` / `treasury` hire a visiting `HERMIT_DOCTOR`!
        - When `affinity >= 55` and `housingCapacity` has room, let a high-`openness` Sect Master / `envoy` invite compatible `wanderingFigures` to join the sect!
        - **Symmetric Counter-Espionage on Campus Visitors (`visitors_clues.js`)**: Give hostile rival sects (`attitude < -45`) a small chance to send a disguised spy as a `'restricted'` guest; if `guard` / `spymaster` / `warden` catches them trespassing near `scripture` or `ANCHOR_*`, capture them into `W.sect.dungeon`!

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4. Mandatory Verification Suite Additions (`08E`)

1. **Fish-Dish Starvation Prevention & Single-Meal Test**: Set `stock.meal = 0, stock.grain = 0`, add `2` cooked fish dishes in `W.fishing`, and assert `A.eat.can(npc, W) === true` and eating consumes `1` fish dish without double-ticking `'meal'` satiety; verify `studyMult` accelerates `A.study.tick`.
2. **27/27 Unique Legendary Equipment Clue Reachability Test**: Extend `visitors_clues_weapons_suite.js` Section 7 so all **27** `VC.LEGENDARY_EQUIPMENT` items have reachable, completable Clue Templates!
3. **Hermit Doctor Fee & Duplicate Recruitment Guard Test (`OUT-07` & `OUT-20`)**: Assert UI-equivalent `HEAL_DISCIPLE` deducts `120` stones into `W.world.resourcePool.stones`, and calling `recruitWanderingFigure` twice returns `ok: false, reason: 'already-recruited'` on the second call.